SMART INDIA HACKATHON 2026

TITLE PAGE

- **Problem Statement ID –** SIH26136
- **Problem Statement Title-** Startup friendly public procurement mechanism that enables government departments to identify, pilot, procure, and scale innovative solutions from eligible startups
- **Theme-** Miscellaneous
- **PS Category-** Software
- **Team ID-**
- **Team Name-** PRODHUB

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IDEA TITLE

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❖ Proposed Solution (Describe your Idea/Solution/Prototype)

SANYOG (संयोग) — an end-to-end digital pathway that takes a government department from a written problem to a scaled, procured startup solution, in eight governed stages: Challenge → Screening → Evaluation → Sandbox → Contract → Milestones → Validation → Scale-up.

❖ Detailed explanation of the proposed solution

- Departments publish outcome-based challenges with a mandatory measured baseline → target; DPIIT-recognised startups are auto-screened for eligibility on application.
- Every pilot runs in a governed sandbox (security clearance, fixed exit date) and pays 30/40/30 against verified milestones, with the MSMED 45-day payment clock visible on each tranche.
- Independent validators record measurements against pre-locked KPIs; three satisfactory departmental endorsements unlock scale-up and a ready GeM listing export.

❖ How it addresses the problem

- Removes turnover, prior-experience and EMD barriers (GFR Rule 173(i) & 170(i) applied as defaults) — while quality criteria are never waived.
- Gives departments the three standard templates — Problem Statement, Pilot Agreement/MOU (IP, data, liability) and Scale-Up Scorecard — as working digital forms, not documents.
- Makes departmental demand visible, payments timed, evaluations accountable and every decision evidence-based and audit-traceable.

❖ Innovation and uniqueness of the solution

- Compliance is computed, not promised — the database itself refuses payment without verification, an active sandbox without security clearance, and rejection without a stated reason.
- KPIs are pre-registered and locked before measurement begins; deemed-approval clocks stop applications from dying in queues; panel scores are attributable with mandatory rationale.
- A validated pilot's evidence pack exports directly to GeM — scale-up across departments without a fresh tender.

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TECHNICAL APPROACH

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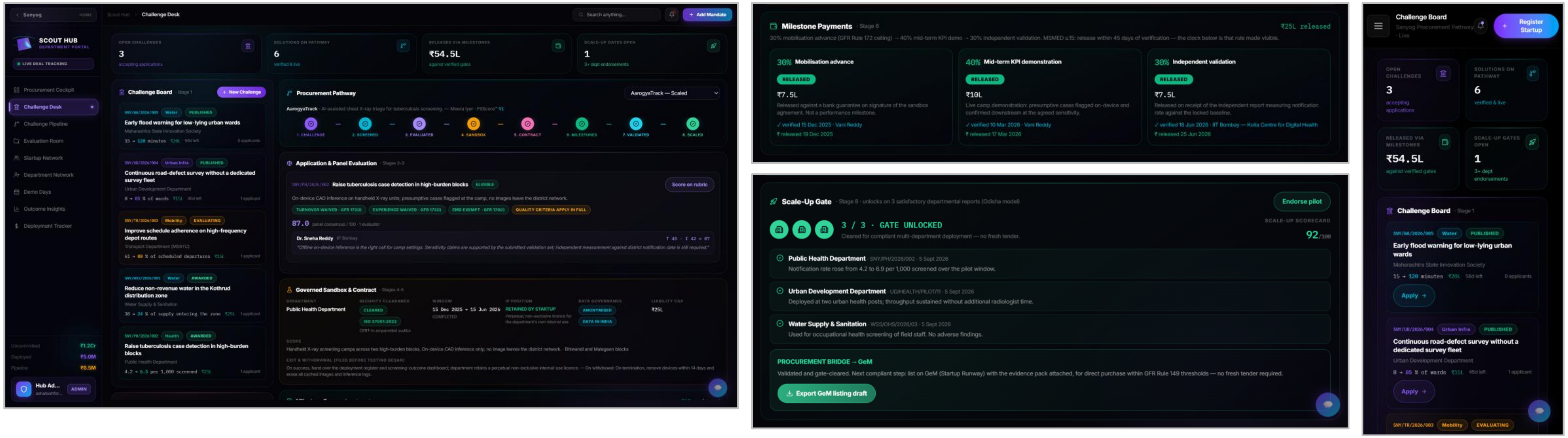
❖ Technologies to be used (e.g. programming languages, frameworks, hardware)

- React 19 + TanStack Start · Cloudflare Workers edge SSR · Supabase PostgreSQL (RLS, column grants) · Groq LLM (FitScore™ + assistant)
- JWT + Google OAuth role-gated portals · DPIIT auto-screening · GeM (Startup Runway) listing export · integrity rules enforced in the database

❖ Methodology and process for implementation (Flow Charts/Images/ working prototype)

1. Challenge published → **2. Eligibility screen** → **3. Panel evaluation** → **4. Sandbox clearance** → **5. Milestone payments** → **6. Independent validation** → **7. Three-dept gate** → **8. GeM scale-up**



Working prototype — live at sanyog.ashutosh-palai2005.workers.dev: Challenge Desk · milestone engine · 3-department scale-up gate with GeM export · mobile view

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FEASIBILITY AND VIABILITY

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❖ Analysis of the feasibility of the idea

- Needs no new law — GFR 2017 already permits every relaxation applied (Rule 173(i) turnover & experience, Rule 170(i) EMD, Rule 172 capping the 30% advance); MSMED s.15/16 supplies the payment clock.
- Policy demand already exists: Maharashtra Innovation Policy 2025, Section 13.7, commits to exactly these procurement guidelines — Sanyog is the working mechanism, ready to adopt.
- Precedents already in force: Telangana G.O. Ms. 08 (deemed approval), Odisha's 3-client scale gate, Kerala GO(Ms) 2/2022 (₹50L direct route). Prototype is fully built and running — not a concept.

❖ Potential challenges and risks

- Subjective or opaque evaluation · late payments starving startups · citizen-data and security exposure · pilots that never convert to procurement (iDEX: ~8% conversion).

❖ Strategies for overcoming these challenges

- Dual-axis 0–50 scores stored per evaluator with a mandatory written rationale and conflict declaration — every decision defensible on audit.
- A 45-day clock on every tranche with overdue MSMED interest (3x bank rate) shown on the department's own screen; release is blocked until verification is recorded.
- CERT-In-empanelled audit before any go-live; ISO/IEC 27001:2022 editions only; data anonymised and localised in India; every sandbox has a fixed exit date.
- The three-department validation gate converts pilot evidence into a GeM listing within GFR Rule 149 thresholds — closing the pilot-to-purchase gap without a fresh tender.

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IMPACT AND BENEFITS

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SIH

❖ Potential impact on the target audience

- Startups — paid within 45 days instead of months; no turnover / experience / EMD walls; departmental demand finally visible; one validated pilot opens every department and district.
- Departments — risk bounded by design (sandbox exit dates, liability caps, IP clarity); evaluation and payment decisions carry evidence; standard templates replace months of drafting.
- Citizens — water, health and mobility solutions reach the field in ~16 weeks, with outcomes measured against locked baselines rather than claimed in reports.

❖ Benefits of the solution (social, economic, environmental, etc.)

- Economic — startup cashflow survives government sales (30% mobilisation advance + clocked tranches); wider MSME participation in public procurement; evidence-based spending.
- Social — transparent, reasoned public decisions with arbitrariness engineered out; successful pilots scale to every district, not just the pilot ward.
- Environmental — pilots target measurable resource loss: non-revenue water cut from 38% toward 24% in the demo arc; transformer failures pre-empted before outages waste supply.
- Governance — 100% of decisions attributable and audit-traceable; scale-up rests on independently validated outcomes, not vendor claims.

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RESEARCH AND REFERENCES

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❖ Details / Links of the reference and research work

- General Financial Rules 2017, Dept of Expenditure — Rules 149 (GeM thresholds), 170(i) (EMD exemption), 172 (30% advance cap), 173(i) (turnover & experience relaxation) · doe.gov.in
- MSMED Act 2006 — s.15 (45-day payment ceiling that cannot be contracted around) and s.16 (default interest at 3x the RBI bank rate, compounded monthly)
- Maharashtra Innovation & Startup Policy 2025 — Section 13.7, startup-procurement guidelines and pilot support · msins.in
- State precedents — Telangana G.O. Ms. No. 08 (deemed approval) · Odisha Startup Policy (3-client scale gate) · Kerala GO(Ms) 2/2022/SPD (₹50L direct purchase)
- CERT-In — Directions 2022 and Guidelines for Government Entities 2023: empanelled pre-hosting audit, 6-hour incident reporting, 180-day logs in India · cert-in.org.in
- DPDP Act 2023 · ISO/IEC 27001:2022 (the 2013 edition expired 31 Oct 2025 — the platform accepts :2022 only)
- iDEX / DISC outcome data, Ministry of Defence — ~548 challenges to ~45 procurement contracts (≈8%): the pilot-to-purchase gap Stage 8 is designed to close
- US SBIR — 15 U.S.C. Section 638(r)(4) · UK SBRI Condition 27 — competitive entry + verified pilot supports a lawful non-competed follow-on; arising IP retained by the supplier
- NHS England KPI framework · HM Treasury Magenta Book — baseline–target–threshold KPI schema and pre-registered evaluation design (adopted in Stages 1 and 7)
- Working prototype (Team PRODHUB) — LIVE at https://sanyog.ashutosh-palai2005.workers.dev · full 8-stage platform with the 3 mandated templates · code: github.com/Ashutoshpalai3073/PRODHUB